

Tensorflow

Keras ??

Tensor $\Rightarrow$ n-dimensional

$\Downarrow$
Matrix

$\Rightarrow$ NN

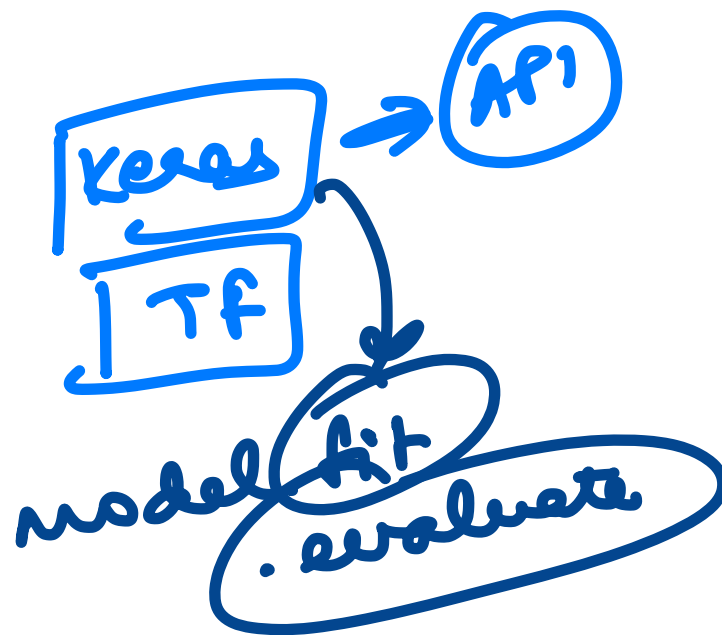
$\Rightarrow$ matrix multiplication

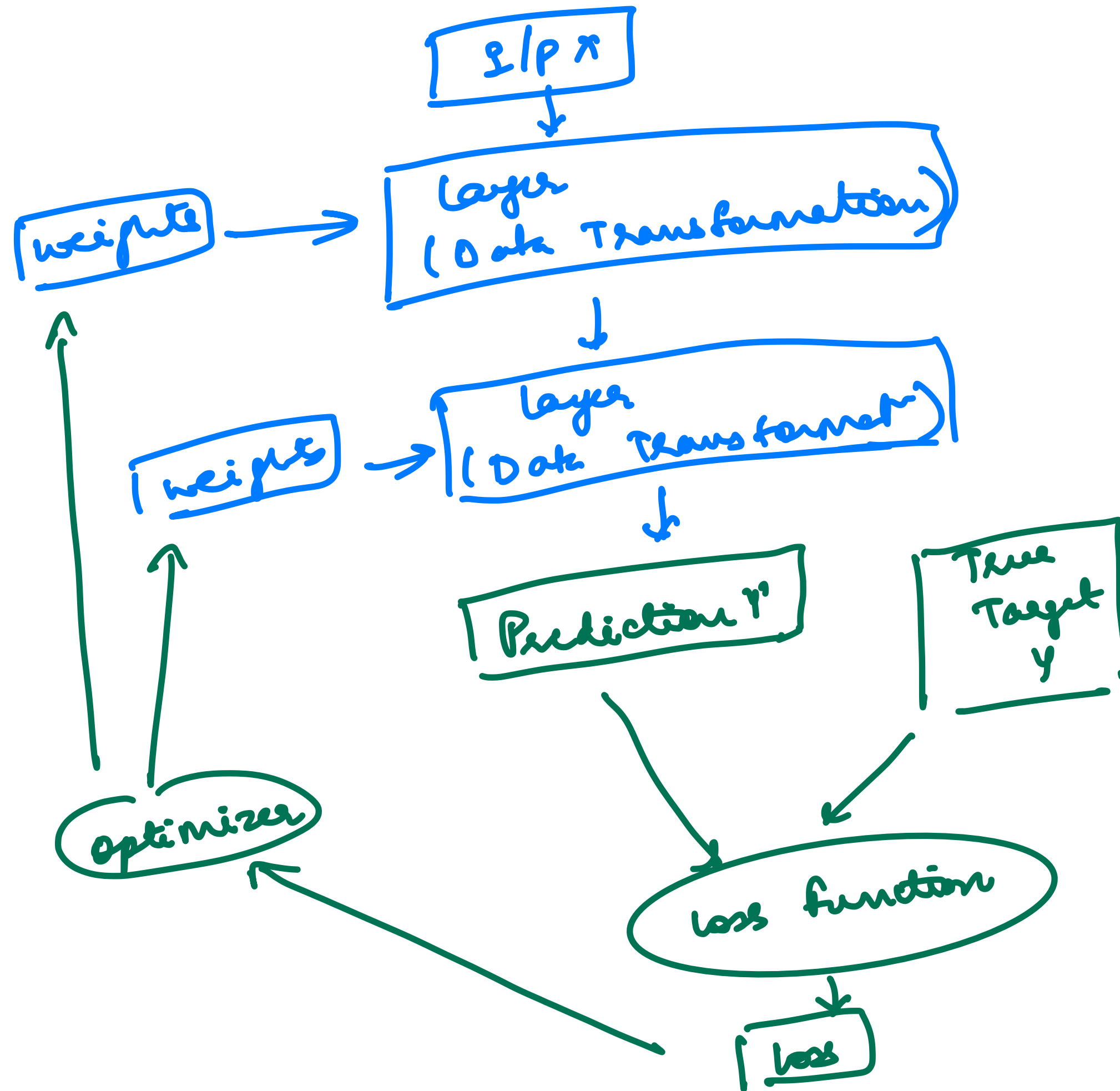
$\Rightarrow$ Gradient
Calculation



NumPy $\rightarrow$ Numerical
Python

TPU





House Price Prediction

① No of bedrooms

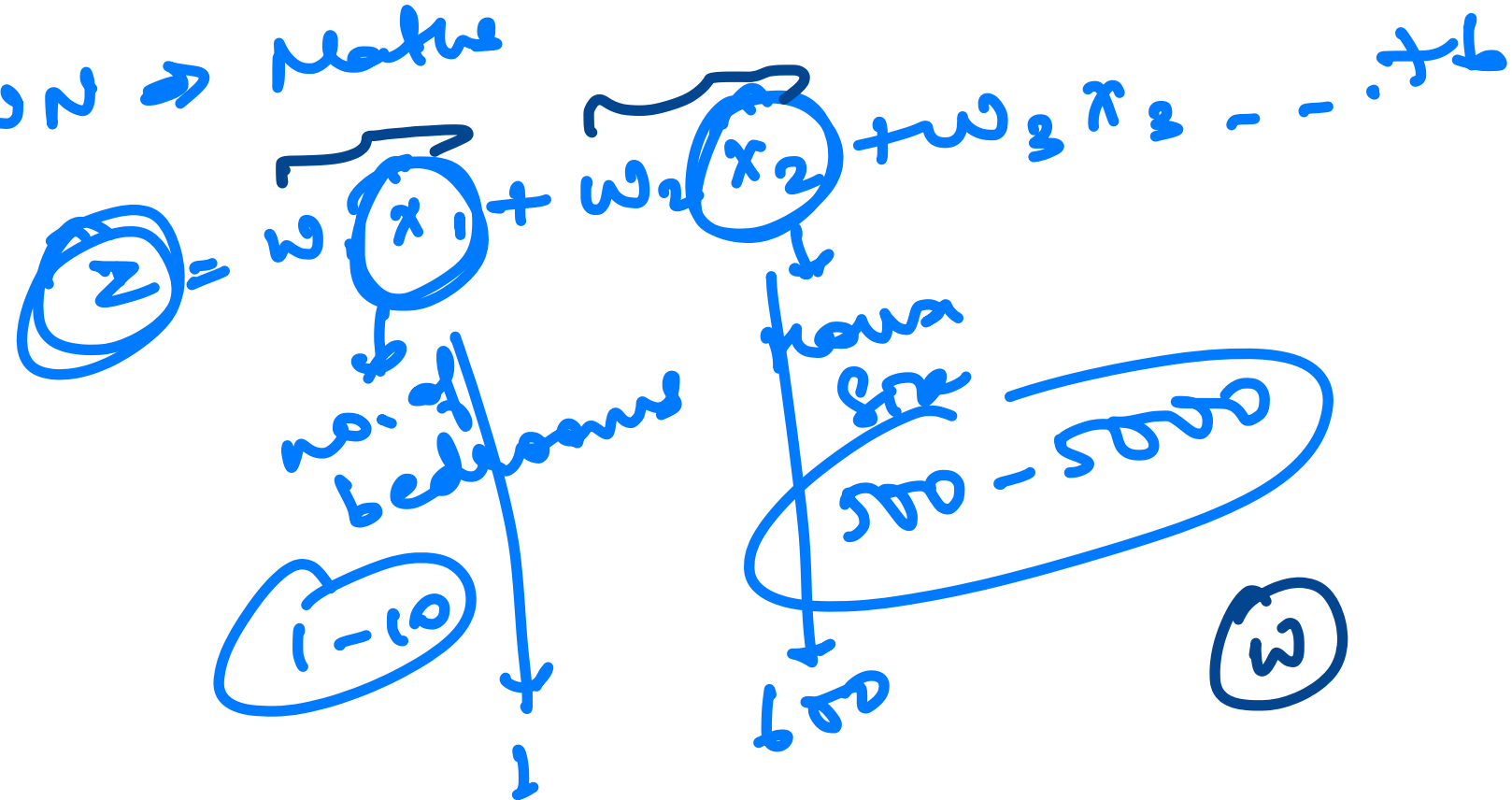
1-10 →

500 - 5000 ✓

0-1
0-1

② House Size

NN → Maths



Activation Function

AF (2)

Sigmoid, Tanh,



$$\frac{1}{1 + e^{-x}}$$

$$\sigma(500) \approx 1$$

$$\sigma(500,000) \approx 1$$

0-1

features
in different
units
height $\rightarrow$ cm
weight $\rightarrow$ kg

0-1 $\rightarrow$ ReLU
-1-1
-1,1
Tanh

$$\frac{0-255}{255} \Rightarrow 0-1 \text{ range}$$

Data Type ✓
Model ✓

Normalization
②

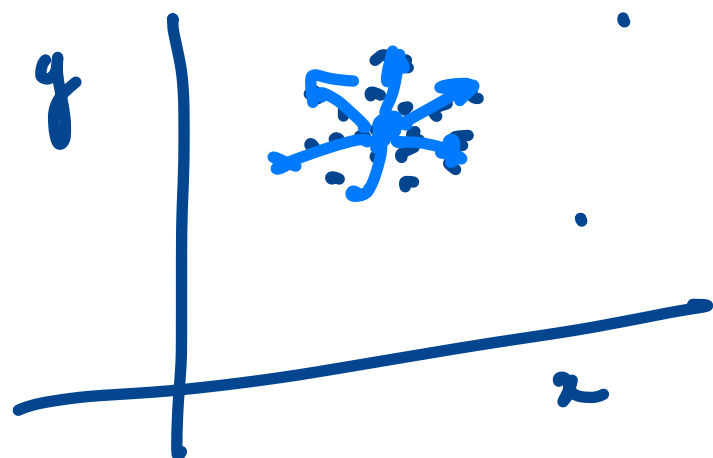
0 → 1
-1 → 1

10, 20, 30, 40, 1500

Outlier

Data point that is
v. different from rest
of your data

0.0...
0-1



mean → avg (μ)
standard deviation → σ

variance → σ²

$$\frac{1}{n} \sum (x - \mu)^2$$

$$\boxed{X_{\text{scaled}}} = \frac{X - \mu}{\sigma}$$

Outliers

Min Max
Scaling

→ Range (0,1)
(-1,1)

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

→ Not good → outliers

→ Data → Range
0-255

Temp → 0-100 → 0-1

→ Sensitive to outliers

2-Score
Standardization

$$X' = \frac{X - \mu}{\sigma} \Rightarrow \begin{matrix} \text{Mean} = 0 \\ \text{Std} = 1 \end{matrix}$$

→ outliers can be handled

Pre-Processing

Text

- lowercase
- Tokenization
-  same length padding
- Remove stop words
"the" "a" "an"

running
↓
run

Image

Scale to fixed size
0-255 → 0-1

Denoising

Colored → grayscale conversion

Audio

→ Denoising

→ Silence removal

→ Amplitude, frequency

Tabular

→ handle null values

→ remove

→ Transform → Mean
log

→ Normalization

→ Removing noise, duplicates, irrelevant features

→ Dimensionality reduction

I love Educosys $\rightarrow$ Numbers?

① Tokenization

I | love | Educosys

I $\rightarrow$ 1
love $\rightarrow$ 2
Educosys $\rightarrow$ 3

hate $\rightarrow$ 4
fabby $\rightarrow$ 5
like $\rightarrow$ 6

cat - 10,000
dog - 100,000

② Encoding

label encoding
one hot encoding (ONE)

Binary vector
cat, dog, elephant, deer, luc

$\begin{bmatrix} 1, 0, 0, 0 \end{bmatrix}$
 $\begin{bmatrix} 0, 1, 0, 0 \end{bmatrix}$
 $\begin{bmatrix} 0, 0, 1, 0 \end{bmatrix}$
 $\begin{bmatrix} 0, 0, 0, 1 \end{bmatrix}$

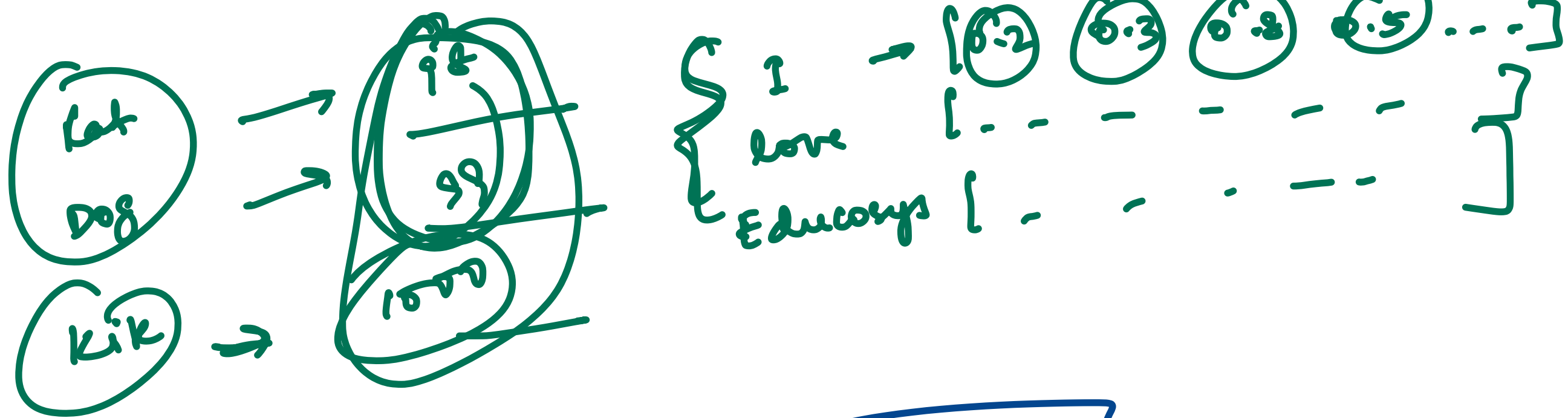
no. of classes
= no. of binary
0/1 in vectors
length of vector
= no. of classes

③ Embedding

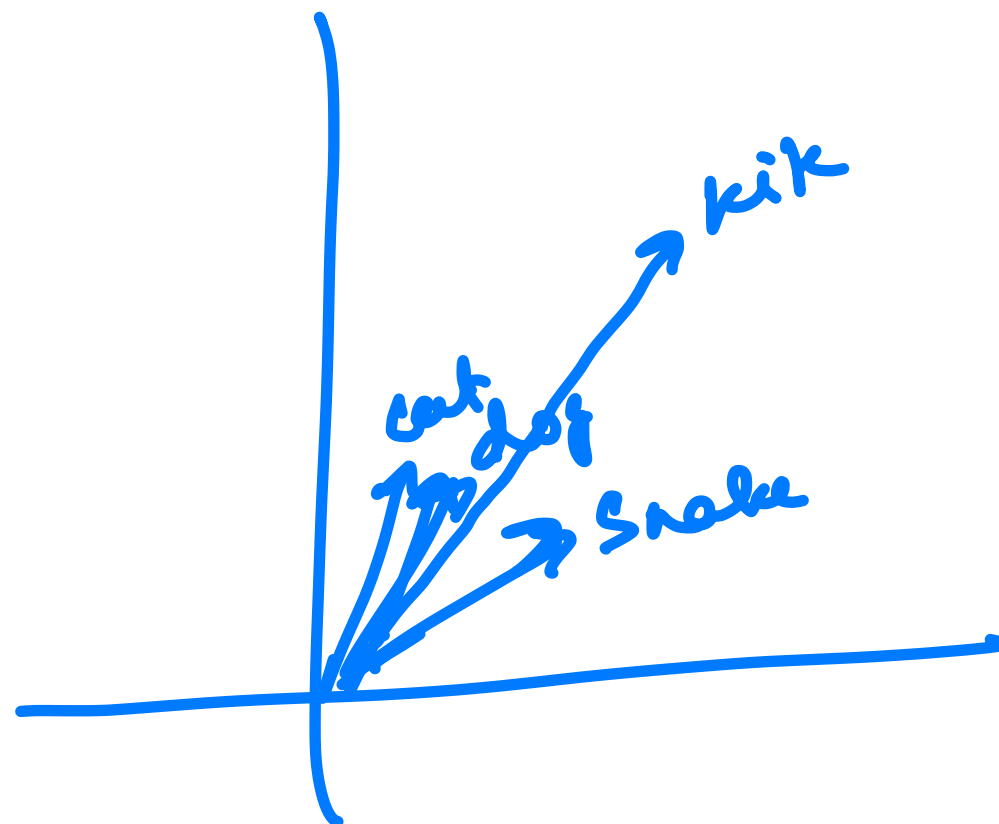
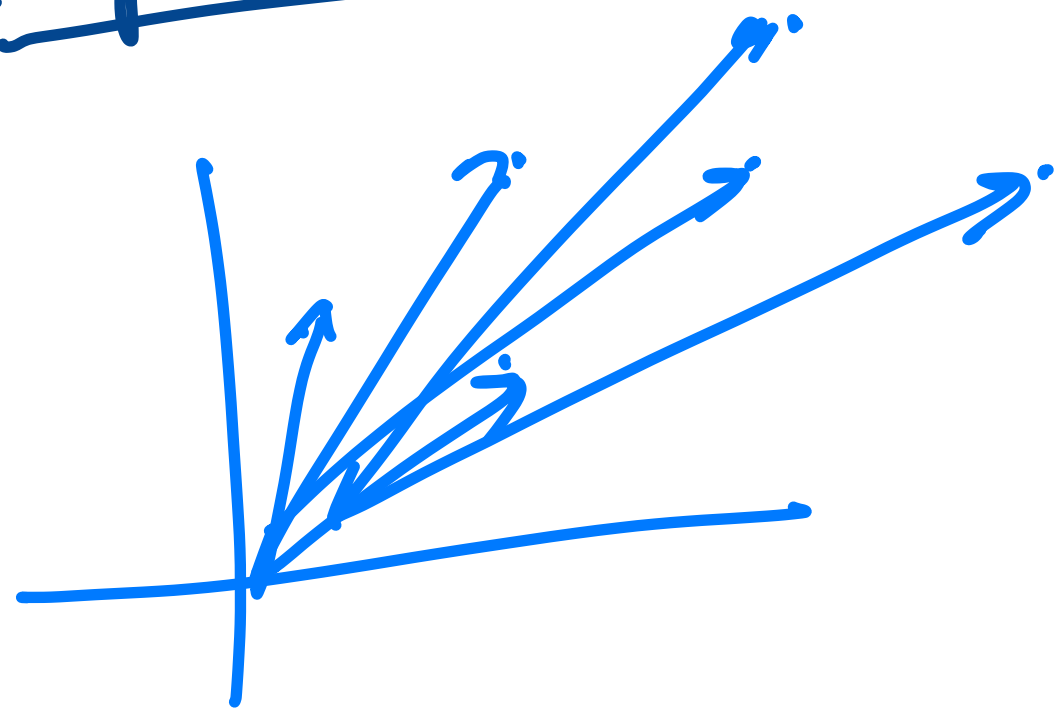
learn context, relationship
between words

Dense

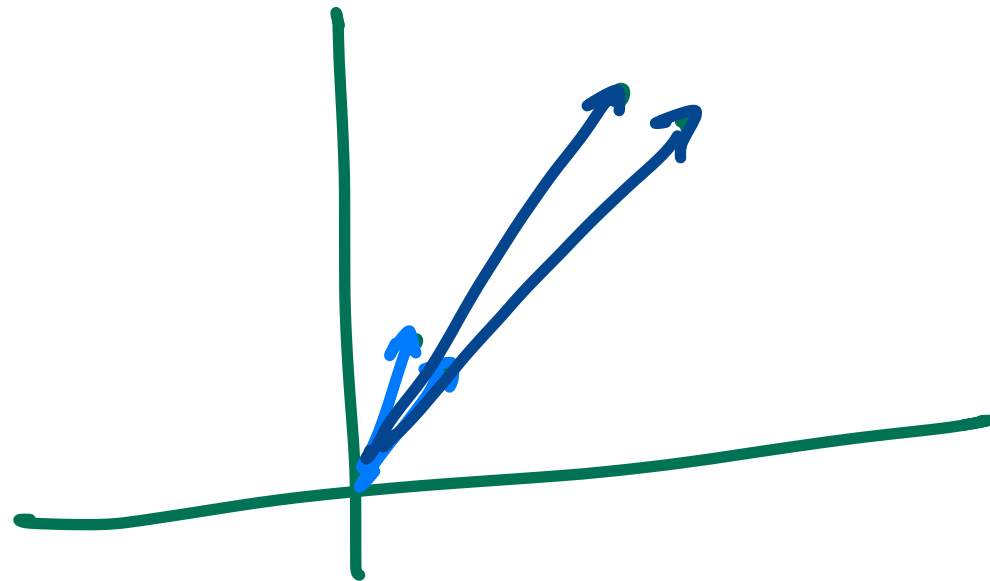
word 2 vec
 $\Downarrow$
neural network



King - man + woman = Queen



$$\begin{array}{l}
 \text{cat} \\
 \text{dog} \\
 \text{hand} \\
 \text{leg}
 \end{array}
 \left\{
 \begin{array}{l}
 \begin{bmatrix} 0.2 & 0.3 & 0.5 & 0.1 & 0.6 \end{bmatrix} \\
 \begin{bmatrix} 0.3 & 0.2 & 0.4 & 0.2 & 0.7 \end{bmatrix} \\
 \begin{bmatrix} 0.7 & 0.8 & 1.0 & -0.5 & 0.4 \end{bmatrix} \\
 \begin{bmatrix} 0.8 & 0.9 & 0.9 & -0.3 & 0.7 \end{bmatrix}
 \end{array}
 \right\}$$



Accuracy
(Percentage %)

% of correct predictions

$$= \frac{\text{No. of correct predictions}}{\text{Total Predictions}} \times 100$$

$$\frac{10,000}{10,000} \rightarrow 98\%$$

200 are wrong

Accuracy $\rightarrow$ More predictions are correct
 $\uparrow$
Loss $\downarrow \Rightarrow$ Closer to true values

Loss

$\hat{y}$ y
 $\downarrow$
Predicted

Loss function

$$\frac{1}{n} \sum (y - \hat{y})^2$$

Classification

Entropy

Binary → Binary Entropy
→ Categorical
Sparse Entropy

Uncertainty

$\log$

binary between 0 and 1

Regression

MSE

MAE

Dependent on targets

House Price Prediction

500k \$

20k \$

30k \$

Regression

Stock Price Prediction

10-500k

MAE →

5-10

Bad

MAE →

200

error

250 Price

House Price Prediction
500k\$ - 500,000k\$

MAE →

200k\$

Good

2000k\$ → Bad.

2500k\$

Temperature

0° to

40°C

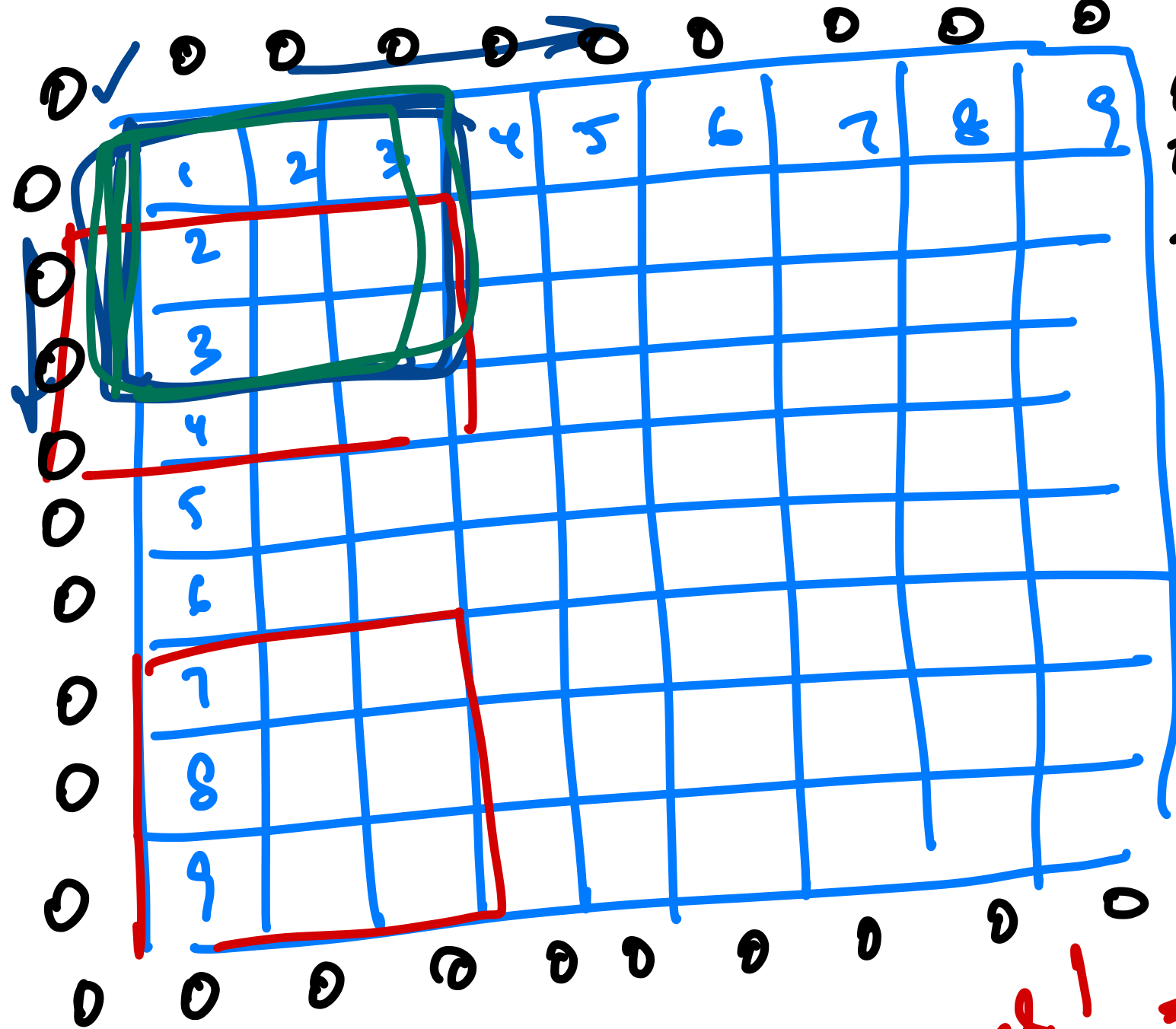
MAE →

3°

2°

→ 10°

→ Bad



- ✓ 1, 2, 3
- ✓ 2, 3, 4
- ✓ 3, 4, 5
- ✓ 4, 5, 6
- ✓ 5, 6, 7
- ✓ 6, 7, 8
- ✓ 7, 8, 9
- 7 = 7

Mode

2/10 → 9 = 9
 0/10 → 7 = 7

9 = 9
 11 = 11

kernel
filter

⇒ 3 = 3

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 1 & 0 & 1 \end{bmatrix}$$

0, 1, 2 3, 4, 5, 6, 7, 8, 9, 10
padding

padding

0, 1, 2 →
1, 2, 3 →
2, 3, 4 →
3, 4, 5 →
4, 5, 6 →
5, 6, 7 →
6, 7, 8 →
7, 8, 9 →
8, 9, 10 →

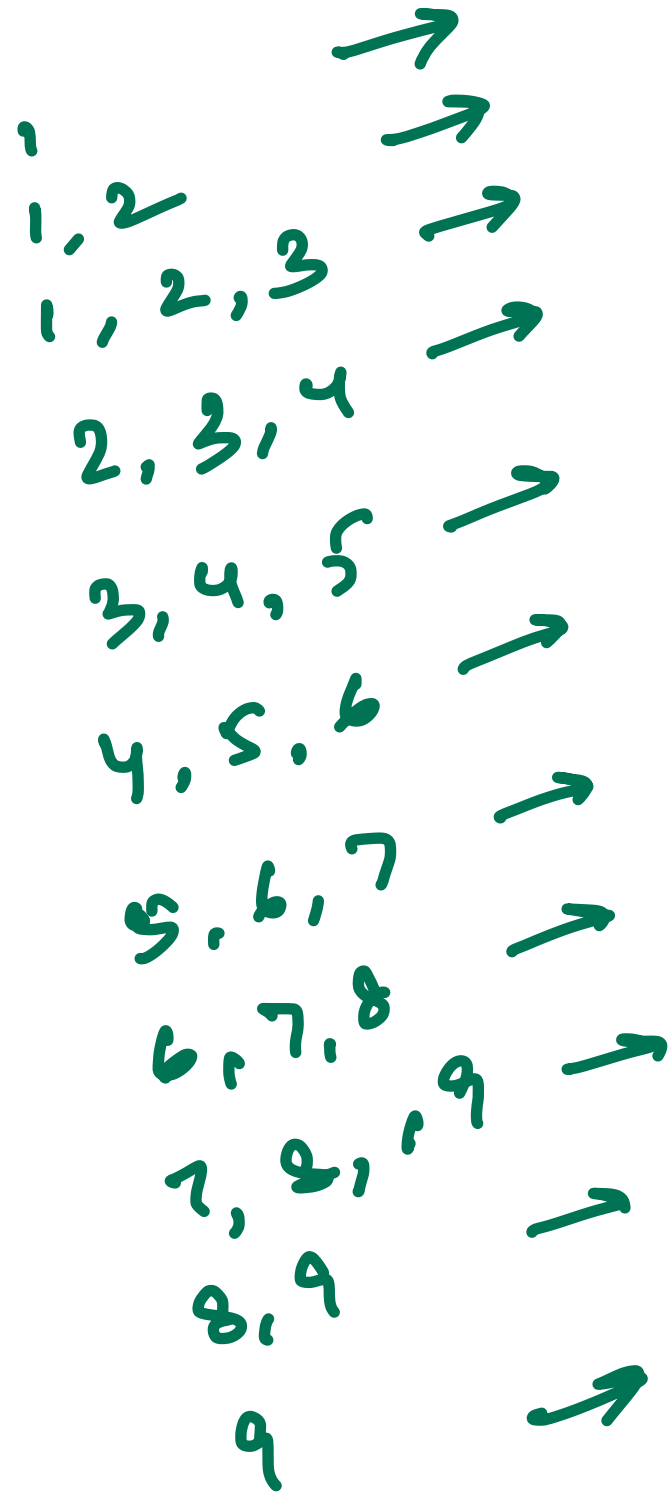
9x9

o/p → same size

as i/p

Add padding across
image.

11 ~ 10



11 ~ 11